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SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

Technical & Tools Declaration

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1. Team & Project Overview

Team Name	LetUsCook
Project / Solution Name	SupplyWise AI
Name (Leader)	Chew Jee Syuen
Email & Phone (Leader)	chewjeesyuen@gmail.com +60 11-2075 2580
University Affiliation(s)	Universiti Malaya (UM)
Live Prototype URL	https://supplywiseai.replit.app
Code Repository URL (public / access granted to judges)	https://github.com/Jeesyuen/LetUsCook

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

SupplyWise AI utilizes a decoupled, high-performance client-first architecture designed for instant responsiveness in kitchen environments. The user interface is built on React 19, TypeScript, and Vite, delivering modular component rendering, strict static type safety, and fast load speeds. State management is orchestrated via Zustand, maintaining synchronized client stores for inventory stock alerts, supplier catalog filtering, and active quotation comparison matrices.

Heavy computation and AI workloads are offloaded to RunPod Serverless / Cloud GPU instances. RunPod hosts our multi-modal LLM parser that ingests unstructured WhatsApp messages, email threads, and PDF price lists, transforming messy text into structured catalog schemas (SKU, price, MOQ, lead time) alongside vector similarity embeddings.

The business logic layer is structured using Python 3.11 and FastAPI, calculating multi-variable landed costs (wholesale rates, Klang Valley road distance freight surcharges, and MOQ gap penalties) while connecting to Orbyte POS models. Supabase (PostgreSQL) is designated for relational supplier records and audit trails in Phase 2. Collaborative development and preview staging are hosted on Replit Core.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
React 19.1.0	Reactive frontend interface and component architecture	MIT
TypeScript 5.8.0	End-to-end static typing and interface contract enforcement	Apache 2.0
Vite 6.2.0	Next-generation frontend tooling and rapid bundling	MIT
Tailwind CSS 4.1.0	Utility-first responsive design tokens and styling	MIT
Zustand 5.0.0	Lightweight reactive client state store and prototype persistence	MIT
Lucide React 1.16.0	Clean, accessible SVG iconography	ISC
FastAPI 0.115.0	High-performance asynchronous backend REST API	MIT
Python 3.11	Computational algorithms and data modeling runtime	PSF

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
RunPod Serverless Inference API	Remote REST endpoint executing LLM unstructured text extraction on WhatsApp messages and quote PDFs	Sponsor-provided ($25 credit)
Supabase REST & Database API (Supabase Inc.)	Cloud BaaS REST API for querying and persisting product catalogs, inventory levels, and purchase orders	Free Tier

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit	Collaborative cloud development environment, CI/CD, and live web preview hosting	Sponsor-provided Core Plan
RunPod Cloud	GPU serverless compute endpoint hosting LLM inference workers	Sponsor-provided / US-IL-1 & US-TX-3
Supabase	Cloud PostgreSQL relational database for catalog persistence and audit trail	Free Tier / ap-southeast-1

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
Qwen/Qwen2.5-7B-Instruct	RunPod Serverless	Pre-trained	Apache License 2.0

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Klang Valley F&B Inventory & Supplier Fixtures	Self-created mock fixtures	Synthetic	N/A	N

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
Antigravity IDE 2.5.5	Code development, documentation refinement	Major
GitHub Copilot	Code completion, UI component logic, and automation script generation	Major
Claude Code	Code development	Major

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

The team affirms that all source code, business logic algorithms, and documentation underwent strict human review, manual validation, and end-to-end architectural testing.

Code assistance tools were utilized strictly for development acceleration and syntax verification under developer supervision. Every data structure, mathematical formula in the landed-cost engine, and user interface component has been compiled, unit-tested, and audited by our team to guarantee zero unverified logic, zero security vulnerabilities, and adherence to production code standards.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
RunPod	$25 Cloud / Serverless Credits	Primary AI compute engine: Hosted the serverless LLM extraction endpoint for unstructured WhatsApp and email supplier quotes.
Replit	Replit Core Subscription	Primary collaborative development workspace, live code sharing, and public prototype staging deployment.

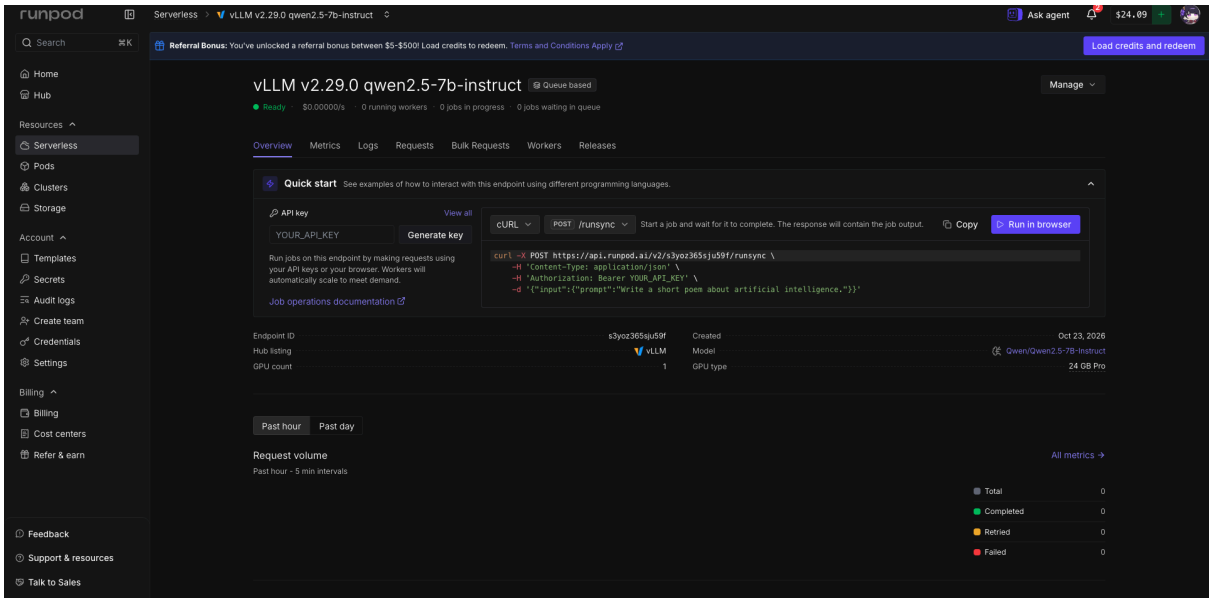
Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

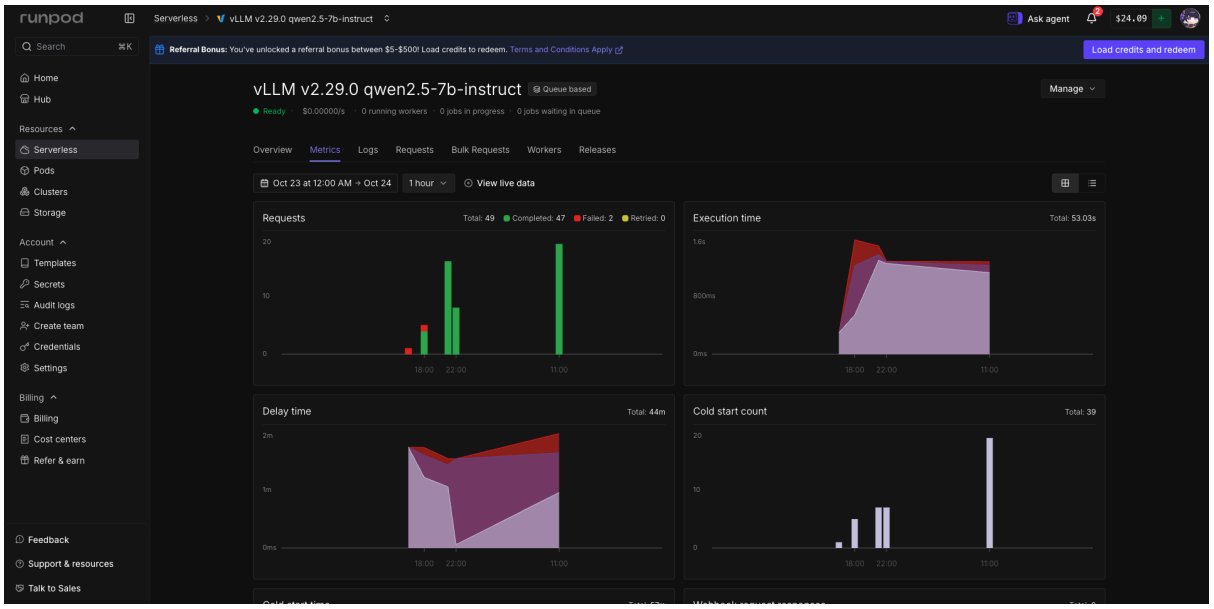
RunPod and Replit were integral to our rapid hackathon development sprint. RunPod provided the specialized GPU infrastructure needed to run low-latency LLM inference without overloading local developer machines, while Replit enabled real-time code collaboration and instant prototype previewing across our five team members.

While the architecture is modular and could theoretically be deployed to alternatives like AWS or self-hosted servers, the sponsor credits were essential for enabling rapid prototyping and low-latency testing within the hackathon timeline.

Proof of use of RunPod:
Endpoint ID: s3yoz365sju59f
GPU type: 24 GB Pro, RTX 4090


runpod
Serverless > vLLM v2.29.0 qwen2.5-7b-instruct
Ask agent
$24.09
Search ⌘K
Referral Bonus: You've unlocked a referral bonus between $5-$500! Load credits to redeem. Terms and Conditions Apply
Load credits and redeem
Home
Hub
Resources
Serverless
Pods
Clusters
Storage
Account
Templates
Secrets
Audit logs
Create team
Credentials
Settings
Billing
Billing
Cost centers
Refer & earn
Feedback
Support & resources
Talk to Sales
vLLM v2.29.0 qwen2.5-7b-instruct
Queue based
Manage
Ready
$0.00000/s
0 running workers
0 jobs in progress
0 jobs waiting in queue
Overview
Metrics
Logs
Requests
Bulk Requests
Workers
Releases
Quick start
See examples of how to interact with this endpoint using different programming languages.
API key
View all
YOUR_API_KEY
Generate key
Run jobs on this endpoint by making requests using your API keys or your browser. Workers will automatically scale to meet demand.
Job operations documentation
cURL
POST /runsync
Start a job and wait for it to complete. The response will contain the job output.
Copy
Run in browser
curl -X POST https://api.runpod.ai/v2/s3yoz365sju59f/runsync \
-H 'Content-Type: application/json' \
-H 'Authorization: Bearer YOUR_API_KEY' \
-d '{"input":{"prompt":"Write a short poem about artificial intelligence."}}'
Endpoint ID
s3yoz365sju59f
Hub listing
vLLM
GPU count
1
Created
Oct 23, 2026
Model
Qwen/Qwen2.5-7B-Instruct
GPU type
24 GB Pro
Past hour
Past day
Request volume
Past hour - 5 min intervals
All metrics
Total 0
Completed 0
Retried 0
Failed 0

Runtime:


runpod
Serverless > vLLM v2.29.0 qwen2.5-7b-instruct
Ask agent
$24.09
Search ⌘K
Referral Bonus: You've unlocked a referral bonus between $5-$500! Load credits to redeem. Terms and Conditions Apply
Load credits and redeem
Home
Hub
Resources
Serverless
Pods
Clusters
Storage
Account
Templates
Secrets
Audit logs
Create team
Credentials
Settings
Billing
Billing
Cost centers
Refer & earn
Feedback
Support & resources
Talk to Sales
vLLM v2.29.0 qwen2.5-7b-instruct
Queue based
Manage
Ready
$0.00000/s
0 running workers
0 jobs in progress
0 jobs waiting in queue
Overview
Metrics
Logs
Requests
Bulk Requests
Workers
Releases
Oct 23 at 12:00 AM → Oct 24
1 hour
View live data
Requests
Total: 49
Completed: 47
Failed: 2
Retried: 0
Execution time
Total: 53.03s
Delay time
Total: 44m
Cold start count
Total: 39

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

No user data is collected

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

Identified Risk / Limitation	Potential Impact	Engineering Mitigation Strategy
Wholesale Price Volatility	Outdated supplier catalog pricing leading to quotation discrepancies.	Implemented dynamic quotation expiration timestamps with automated freshness alerts prompting re-verification.
Network Latency in Kitchens	Unstable Wi-Fi causing delayed stock updates during peak hours.	Integrated optimistic UI state updates and client-side Zustand persistence for uninterrupted local operation.
Delivery Surcharge Spikes	Unmet supplier MOQs leading to unexpectedly high landed unit costs.	Built real-time MOQ shortfall indicators and visual delivery fee warnings with suggested bundle additions.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☑ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☑ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☑ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☑ Any use of official SDC sponsor resources has been declared in Section 5.
- ☑ The prototype described here is accessible to judges through the links provided in Section 1.
- ☑ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.



25 Oct 2026